Novozymes Enzyme Stability Prediction

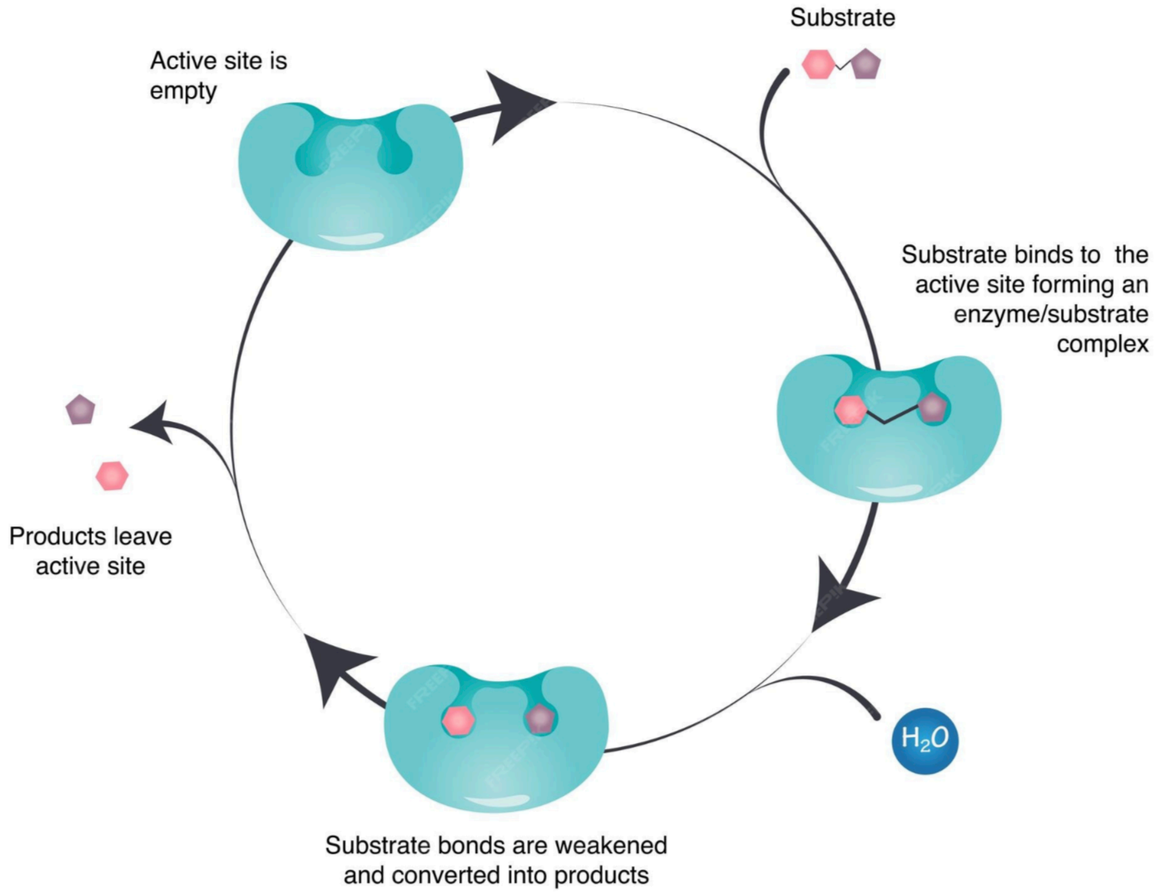
Kaggle Machine Learning Competition: 24th Silver Medal Solution; Joint work with Haoyuan Liu (Dynavax Technologies, UCLA)

Presenter: Chuanwan (Christine) Wu

Introduction

Enzymes & their functionality

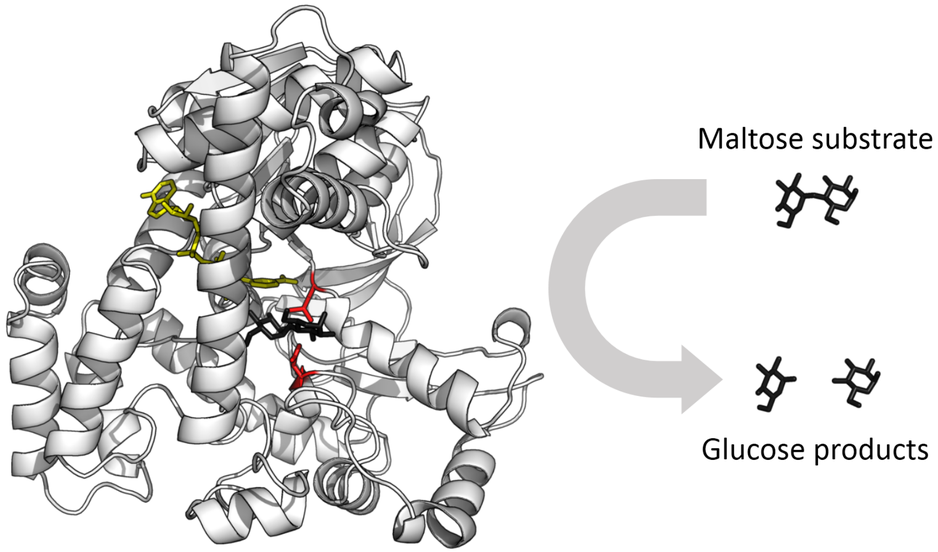
Enzymes are proteins that act as biological catalysts, speeding up chemical reactions in living organisms without being consumed in the process.



The catalytic cycle of an enzyme.

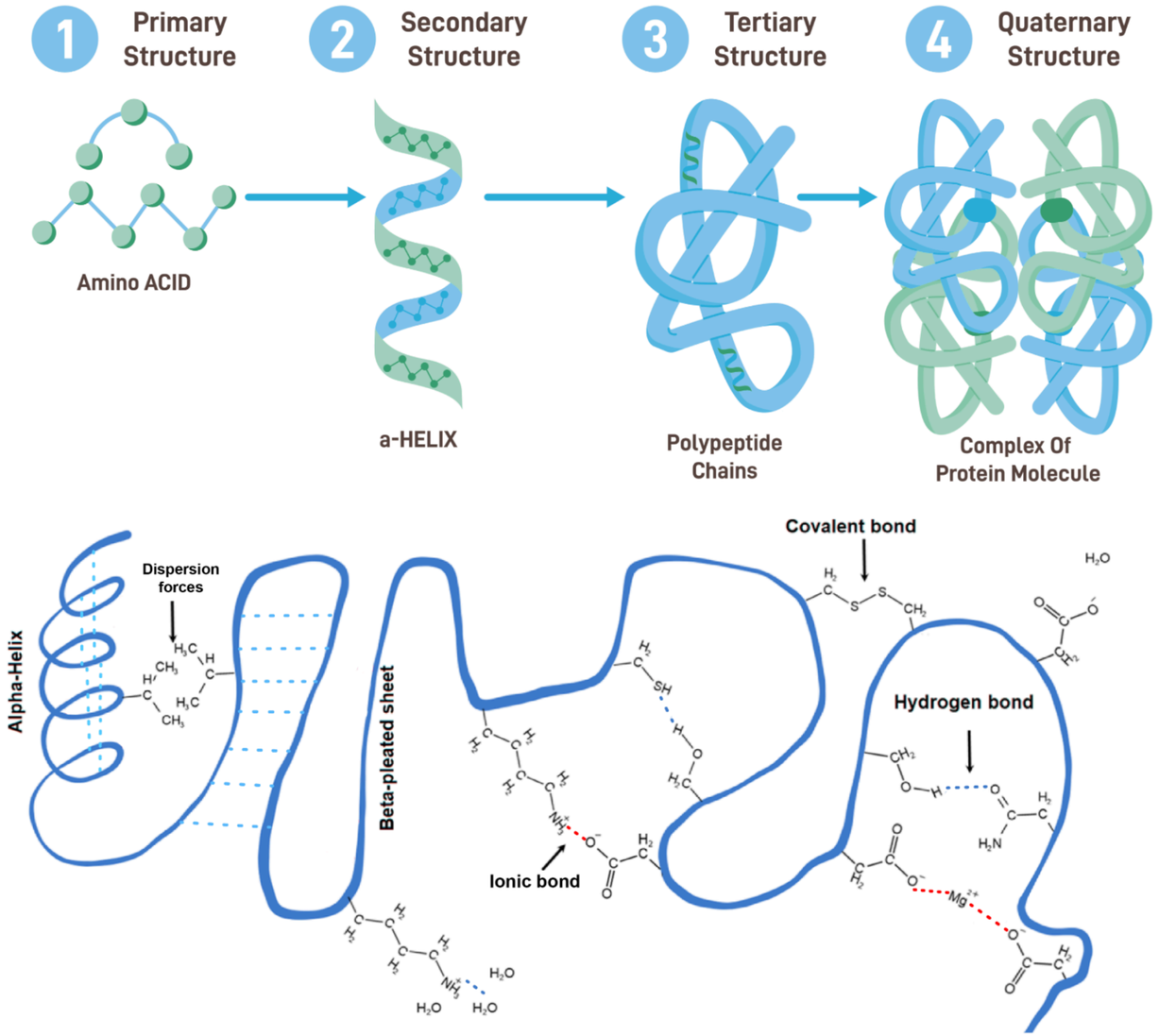
Wikipedia contributors. (n.d.). Enzyme. In Wikipedia. Retrieved May 23, 2024, from *https://en.wikipedia.org/wiki/Enzyme*

Freepik. (n.d.). Vector illustration of a diagram of an enzyme catalytic cycle template [Image]. Freepik. Retrieved May 23, 2024, from *https://www.freepik.com/premium-vector/vector-illustration-diagram-enzyme-catalytic-cycle-template_30160592.htm*

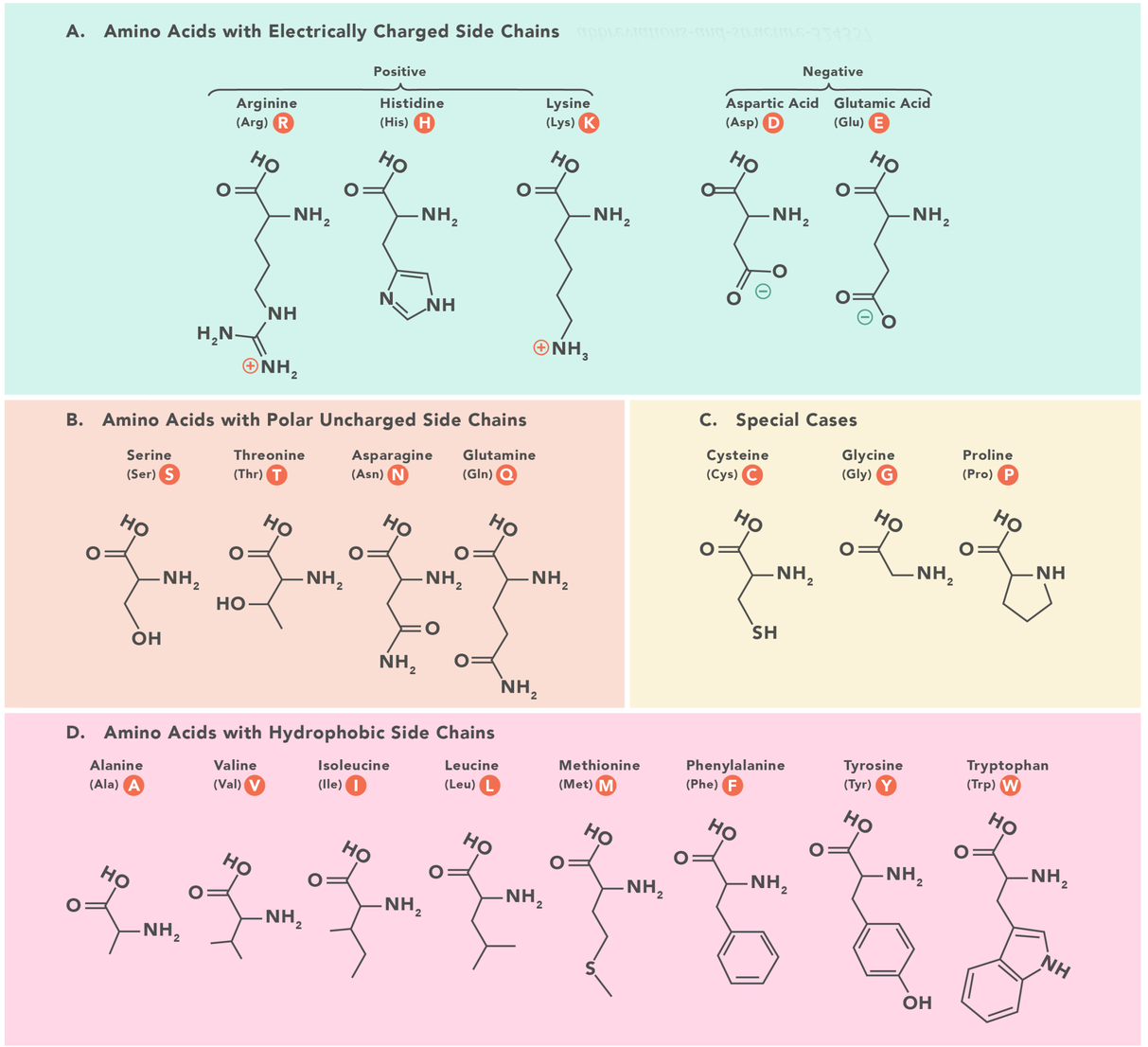


The enzyme *glucosidase* converts the sugar maltose into two glucose sugars. Active site residues in red, maltose substrate in black, and NAD cofactor in yellow.

Enzymes as proteins *



Protein structural levels (top); molecular interactions on amino acids (bottom).



Amino Acid chart.

Monashuniversity. (n.d.). Protein structure. Student Academic Success. Retrieved May 23, 2024, from *https://www.monash.edu/student-academic-success/chemistry/medicinal-chemistry/protein-structure*

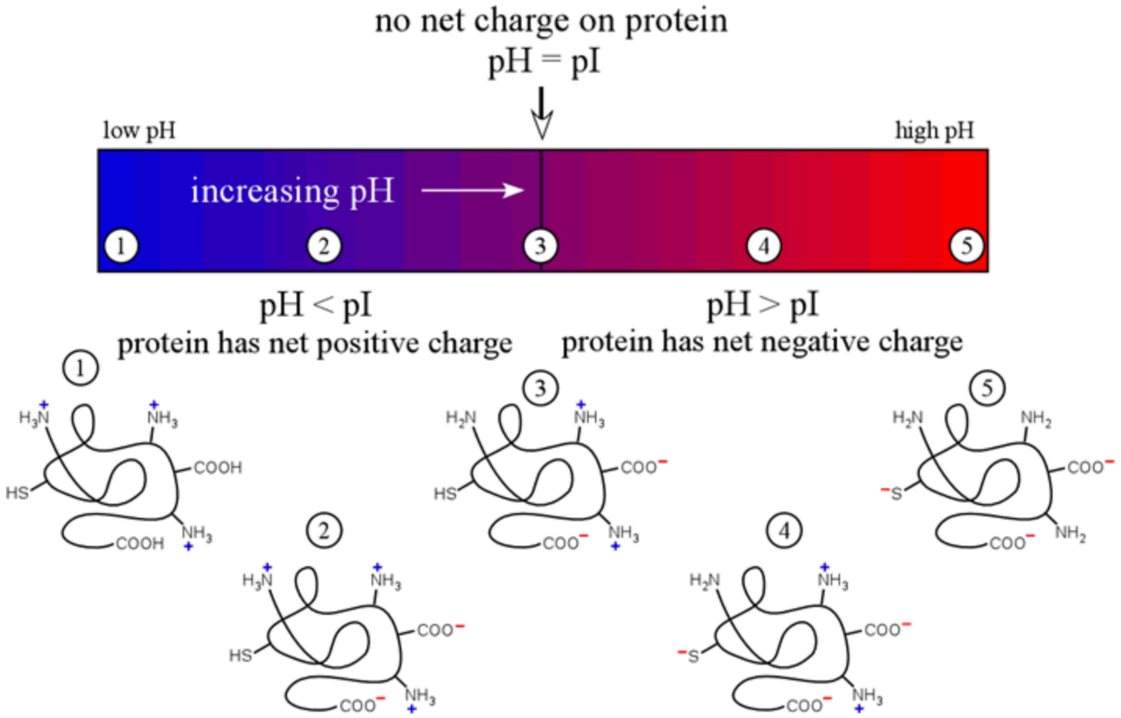
Steward, K., PhD. (2023, December 18). *Essential Amino Acids: Chart, Abbreviations and structure*. Applied Sciences From Technology Networks. *https://www.technologynetworks.com/applied-sciences/articles/essential-amino-acids-chart-abbreviations-and-structure-324357*

Proteins, pH & thermostability **

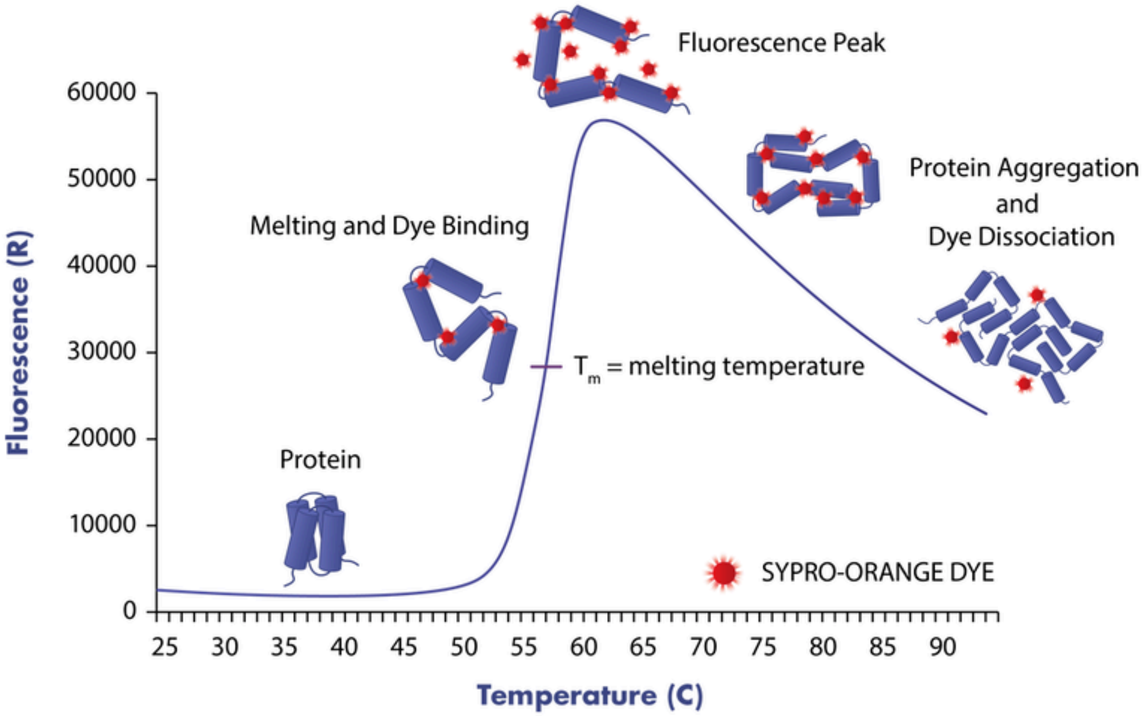
Cronk, J. D. (n.d.). Protein methods. Gonzaga University. Retrieved May 23, 2024, from *http://guweb2.gonzaga.edu/faculty/cronk/CHEM245pub/protein_methods.html*

Reaction Biology. (n.d.). Thermal Shift Assay (TSA). Retrieved May 23, 2024, from *https://www.reactionbiology.com/services/biophysical-assays/thermal-shift-assay-tsa*

The pH of a solution and temperature can dramatically affect a protein's stability, primarily by altering its state, which can lead to unfolding and loss of functionality.



How the charge on a protein would vary with the "ambient" pH of the medium of which it is a component.

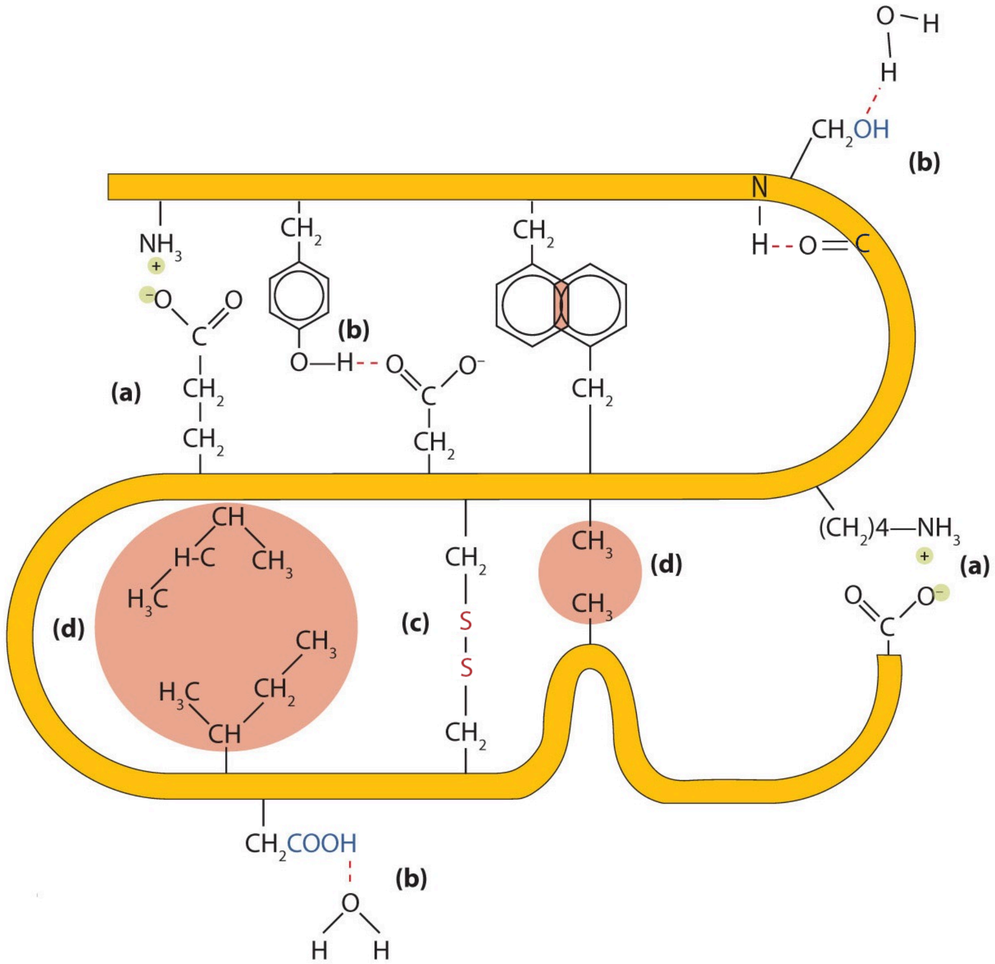


How the temperature can affect a protein's stability.

Proteins, folding & unfolding

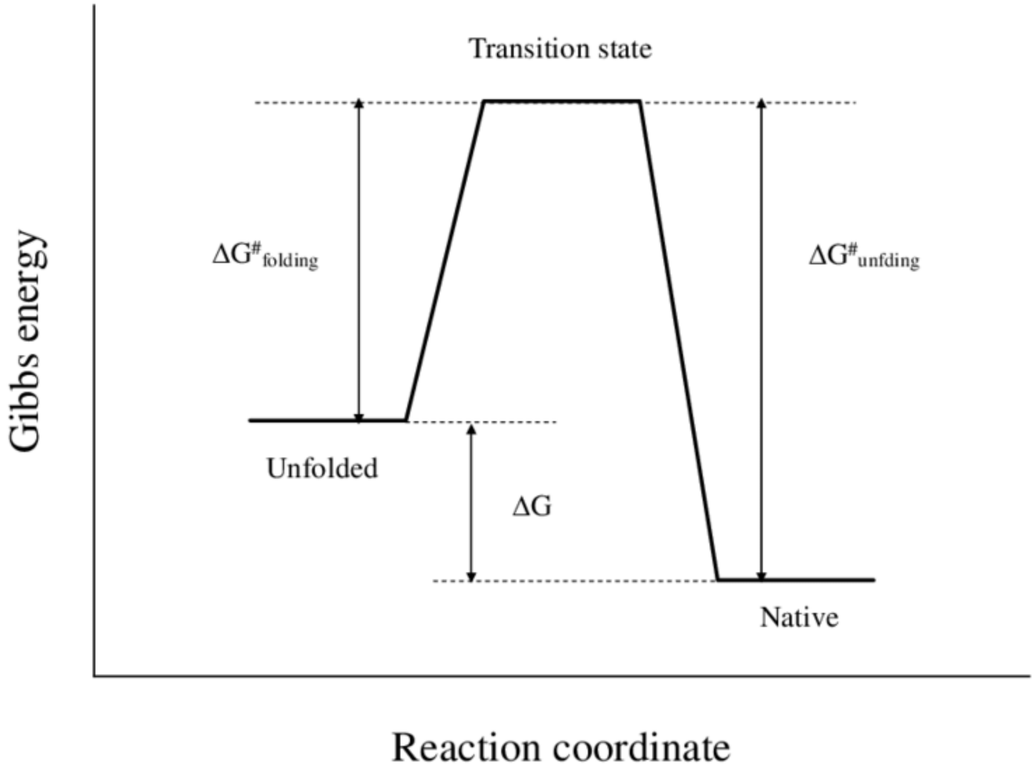
Cannon, J. G. (2015, November 4). 4.17: 4.43 Proteins. In MGA CHEM 1152K Survey of Chemistry II – Introduction to Biochemical Molecules. LibreTexts. https://chem.libretexts.org/Courses/Middle_Georgia_State_University/MGA_CHEM_1152K_Survey_of_Chemistry_II/Chapters/04%3A_Introduction_to_Biochemical_Molecules/4.17%3A_4.43_Proteins

Mukaiyama, A., & Takano, K. (2009). Slow unfolding of monomeric proteins from hyperthermophiles with reversible unfolding. [Figure: Schematic energy diagram of two-state protein folding]. ResearchGate. https://www.researchgate.net/figure/Schematic-representation-of-the-energy-diagram-for-the-protein-folding-process-assuming_fig9_24366289



Tertiary Protein Structure Interactions. Four interactions stabilize the tertiary structure of a protein: (a) ionic bonding, (b) hydrogen bonding, (c) disulfide linkages, and (d) dispersion forces.

Tertiary interactions lower the folded state's Gibbs free energy, and amino acids that strengthen these interactions make ΔG more negative, boosting thermostability.



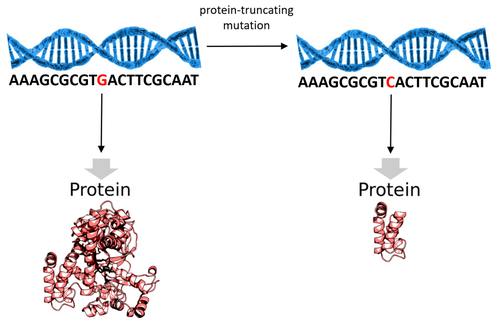
A schematic two-state protein-folding energy diagram showing the Gibbs free-energy activation barriers for folding (ΔG‡ folding) and unfolding (ΔG‡ unfolding).

Proteins & mutations ***

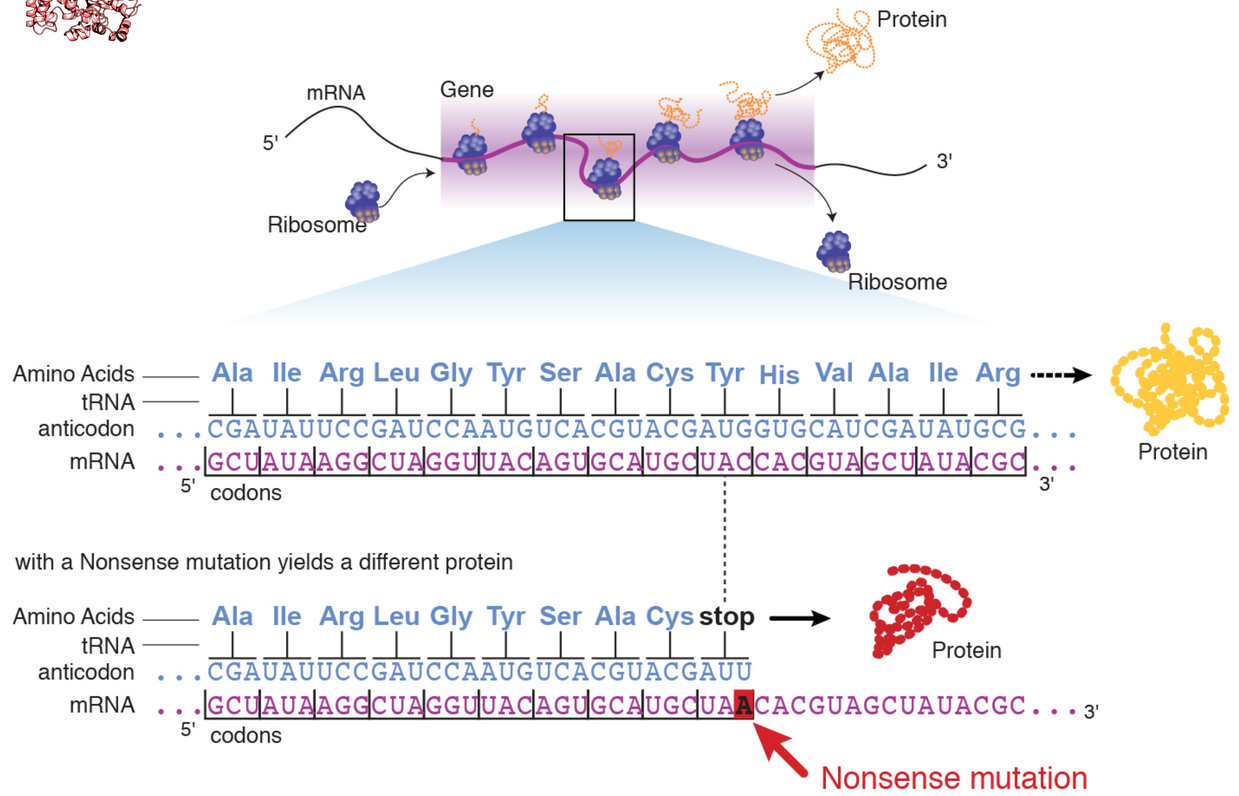
The Dish on Science. (n.d.). [Article Title]. Stanford University. Retrieved May 23, 2024, from *https://thedishonscience.stanford.edu/articles/ekg280kcyxt6r9dwmszsvgr1cxlwpj*

AR Inspired Pencil. (n.d.). DNA mutation deletion [Image]. Retrieved May 23, 2024, from *https://ar.inspiredpencil.com/pictures-2023/dna-mutation-deletion*

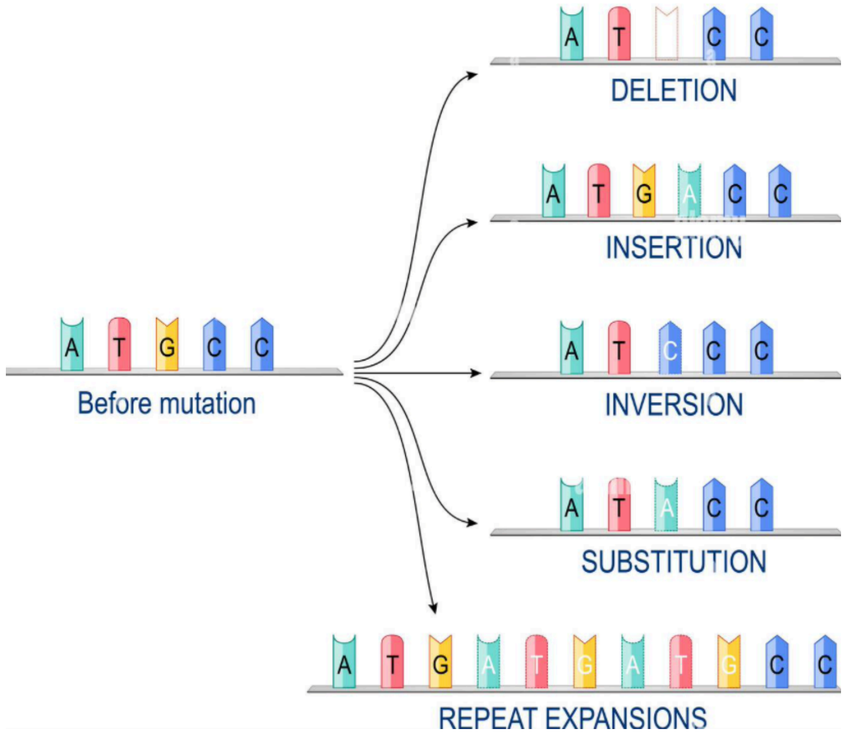
Oncology Nursing Society. (n.d.). Variant. ONS.org. Retrieved May 23, 2024, from *https://onf.ons.org/variant*



A mutation first occurs in the DNA sequence of the gene that codes for the protein. It then passes through mRNA in transcription, and mRNA to protein in translation.



Mutations allow us to redesign natural enzymes (wild-type) to withstand harsh industrial conditions.



Different types of mutations.

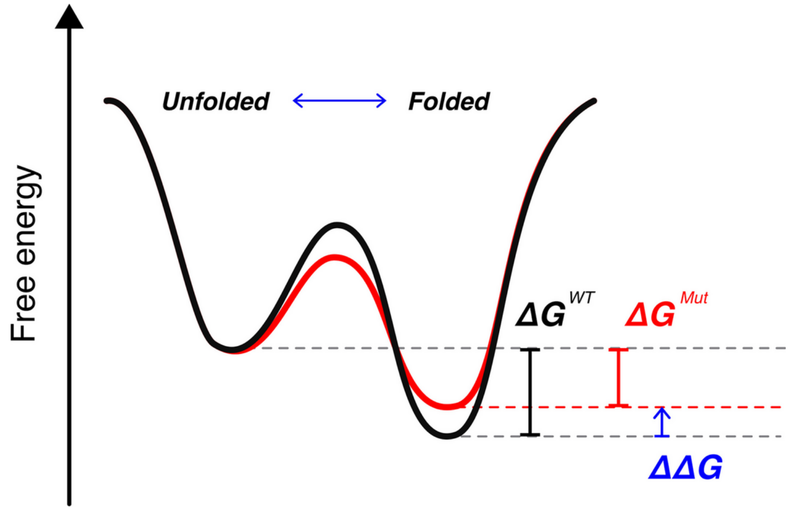
Proteins, folding & Gibbs free energy & thermostability

Ehrlich, H., Rodrigues, J., Camacho, C., & Philippi, S. (2023). Statistical modeling to quantify the uncertainty of FoldX-predicted protein folding and binding stability. [Figure 1: Visualization of protein folding energetics]. ResearchGate. https://www.researchgate.net/publication/375596394_Statistical_modeling_to_quantify_the_uncertainty_of_FoldX-predicted_protein_folding_and_binding_stability

A computational metric that predicts relative melting temperature Tm by evaluating how single-point mutations alter the Gibbs free energy of folding (ΔG) and the resulting stability difference (ΔΔG).

$$\Delta G = -RT\ln\frac{[folded]}{[unfolded]} \qquad (*)$$

$$\Delta\Delta G = \Delta G_{mutant} - \Delta G_{wild-type} = -RT\ln\frac{[folded]_{mutant}/[unfolded]_{mutant}}{[folded]_{wild-type}/[unfolded]_{wild-type}} \qquad (**)$$



Visualization of folding energetics. ΔG shows whether a mutation raises (destabilizes) or lowers (stabilizes) the folded-state energy. The red curve depicts a destabilizing mutation.

ΔΔG of folding correlates with changes in melting temperature (Tm):

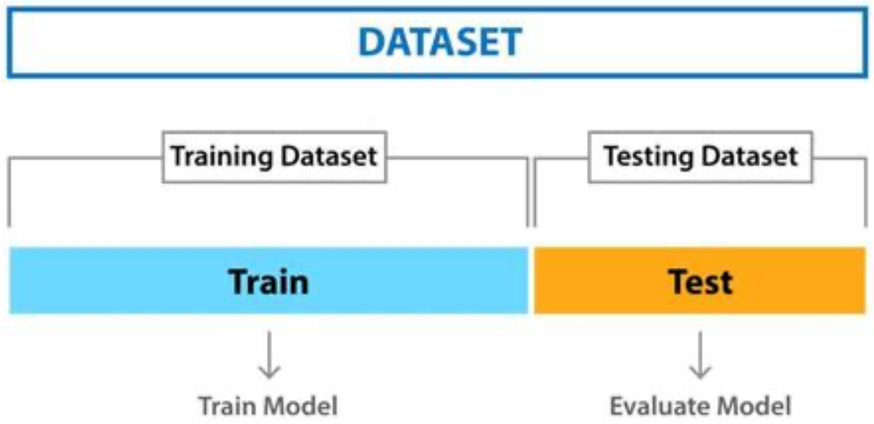
1. Negative ΔΔG → increased stability → higher Tm (more thermostable).
2. Positive ΔΔG → decreased stability → lower Tm (less thermostable).

Objective

Training & testing datasets

VProExpert. (n.d.). Machine learning model validation. Retrieved May 23, 2024, from *https://www.vproexpert.com/machine-learning-model-validation/*

Kaggle. (2022). Novozymes Enzyme Stability Prediction [Data set]. Kaggle. https://www.kaggle.com/competitions/novozymes-enzyme-stability-prediction



i. Protein Sequence (**Enzymes as proteins** *)
ii. pH (**Protein, pH & thermostability** **)
iii. Melting point tm (**Protein, pH & thermostability** **)
iv. Sequence id
v. Data source

i. Single-mutation variant of an enzyme (**Protein & mutations** ***)

Objective:
Predict relative protein thermostability from single amino-acid mutations using only the sequence.

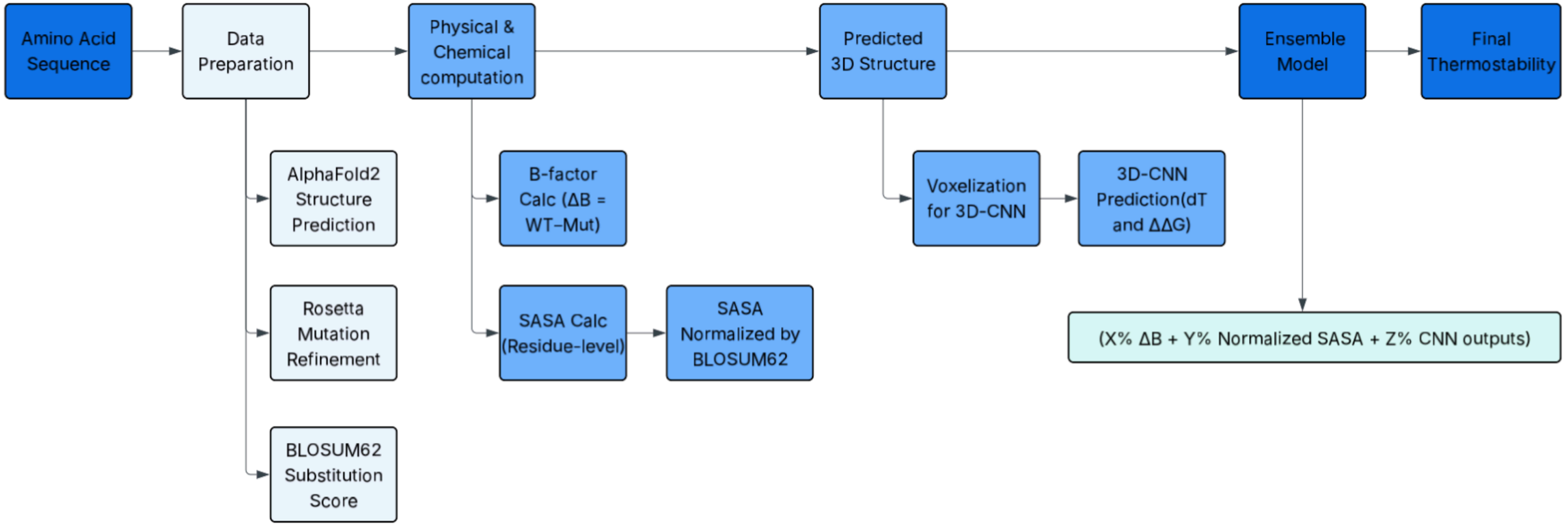
Details:

- The test set contains experimental melting temperatures for 2,413 single-mutation variants of an enzyme (GenBank: KOC15878.1), provided by Novozymes A/S.
- The wild-type amino acid sequence is provided.
- Evaluation metric: Spearman's rank correlation coefficient between experimental and predicted melting temperatures.

Methods

Methodology overview

Process flow chart:



Other models tried in the past:

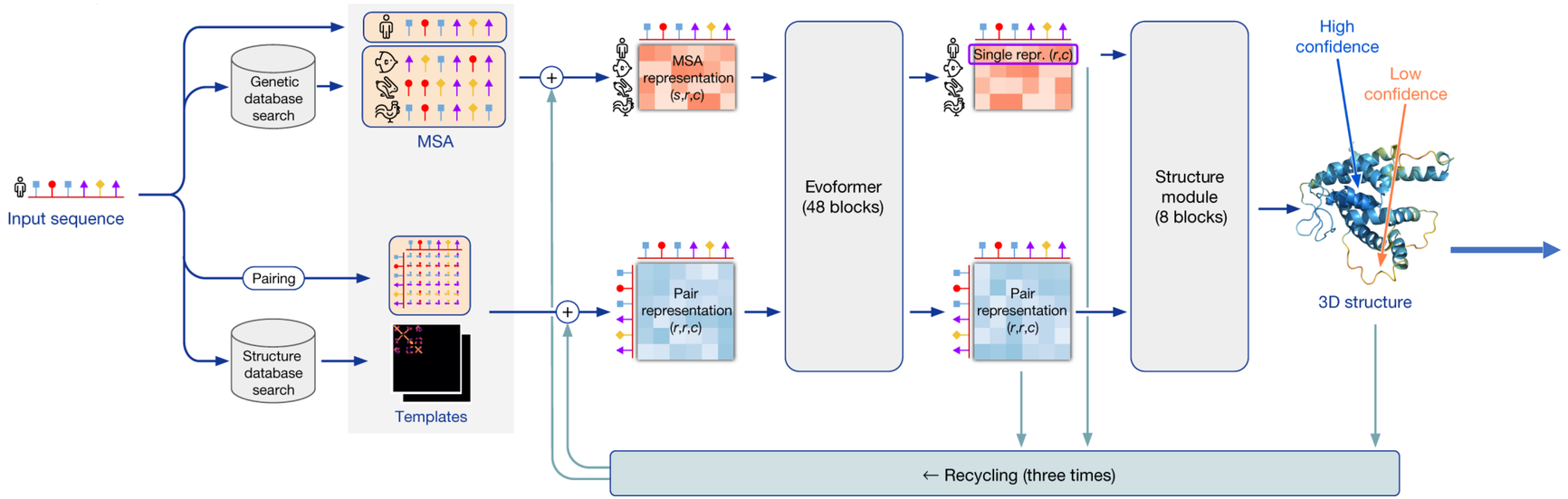
- Extreme Gradient Boosting (XGBoost)
- ESM-2

** Note: The data files were taken from the kaggle community*

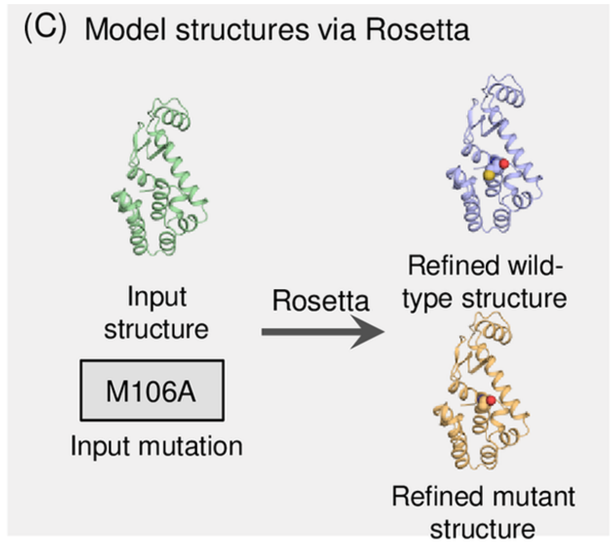
i. Data preparation: Alphafold2 & Rosetta

Jumper, J., Evans, R., Pritzel, A., Green, T., Figurnov, M., Ronneberger, O., et al. (2021). Highly accurate protein structure prediction with AlphaFold. Nature, 596(7873), 583–589. https://doi.org/10.1038/s41586-021-03819-2

Alford, R. F., Leaver-Fay, A., Jeliazkov, J. R., O'Meara, M. J., DiMaio, F. P., Park, H., et al. (2017). The Rosetta all-atom energy function for macromolecular modeling and design. Journal of Chemical Theory and Computation, 13(6), 3031–3048. https://doi.org/10.1021/acs.jctc.7b00125



Protein Structure Prediction via Alphafold2



Mutant Protein Structure Refinement via Rosetta

i. Energy minimization of each mutant using Rosetta's all-atom energy function and the FastRelax protocol
ii. The Rosetta energy function models biomolecular conformational energy as

$$\Delta E_{\text{total}} = \sum_i w_i E_i(\Theta_i, \text{aa}_i)$$

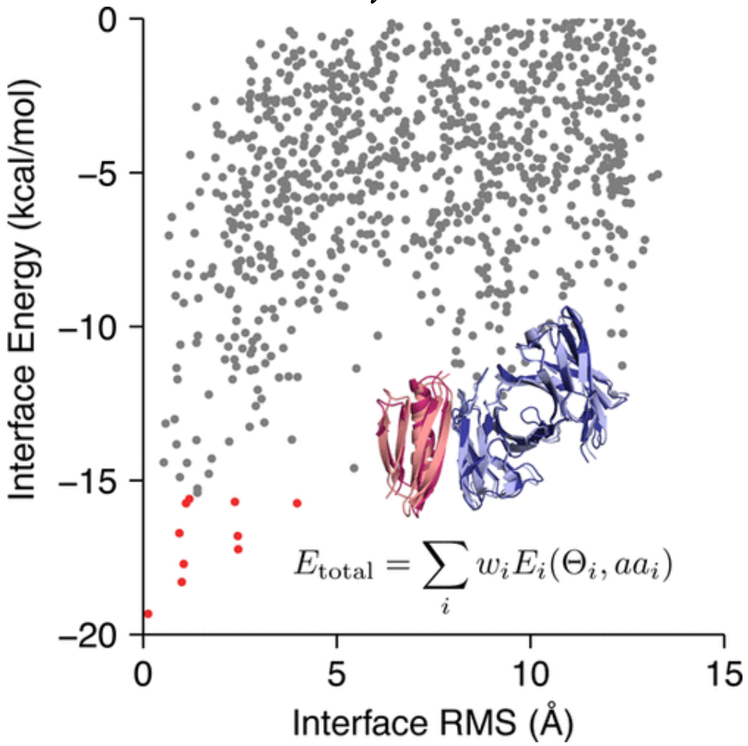
where each term E_i depends on geometric parameters (Θ) and amino-acid identities (aa)

i.i More on Rosetta energy

Alford, R. F., Leaver-Fay, A., Jeliazkov, J. R., O'Meara, M. J., DiMaio, F. P., Park, H., et al. (2017). The Rosetta all-atom energy function for macromolecular modeling and design. Journal of Chemical Theory and Computation, 13(6), 3031–3048. https://doi.org/10.1021/acs.jctc.7b00125

Rosetta's energy function integrates atomic interactions, solvation and electrostatics, torsional and geometric constraints, and reference energies to model protein stability and design.

$$\Delta E_{\text{total}} = \sum_i w_i E_i(\Theta_i, \text{aa}_i)$$



Energy Terms (E_i):

1. Atom-Pair Interactions
   - Hydrogen Bonds
   - Van der Waals
   - Solvation
   - Electrostatics
   - Disulfide Bonding
2. Protein Backbone and Side-Chain Torsions
   - Ramachandran Maps
   - Backbone Design
   - Side-Chain conformations
3. Special-Case Torsions
4. Modeling Non-ideal Bond Lengths and Angles
5. Protein Design (Reference Energy)

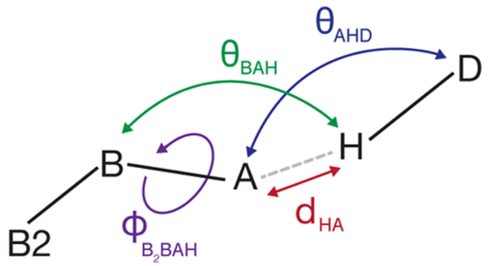
ΔΔG of Mutation to analyze mutants thermostability

i.ii More on Rosetta energy

Alford, R. F., Leaver-Fay, A., Jeliazkov, J. R., O'Meara, M. J., DiMaio, F. P., Park, H., et al. (2017). The Rosetta all-atom energy function for macromolecular modeling and design. Journal of Chemical Theory and Computation, 13(6), 3031–3048. https://doi.org/10.1021/acs.jctc.7b00125

Rosetta's energy on atomic pair interactions.

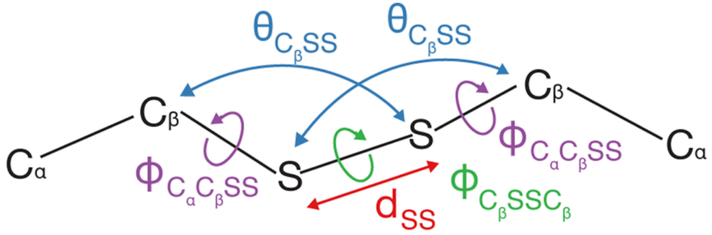
H-boding



A = Acceptor H = Hydrogen
D = Donor B = Base B2 = Parent

$$E_{\text{hbond}} = \sum_{\text{H,A}} w_{\text{H}} w_{\text{A}} f(E^{\text{HA}}_{\text{hbond}}(d_{\text{HA}}) + E^{\text{AHD}}_{\text{hbond}}(\theta_{\text{AHD}}) + E^{\text{BAH}}_{\text{hbond}}(\theta_{\text{BAH}}) + E^{\text{B}_2\text{BAH}}_{\text{hbond}}(\rho, \phi_{\text{B}_2\text{BAH}}, \theta_{\text{BAH}}))$$

Disulfide bonding



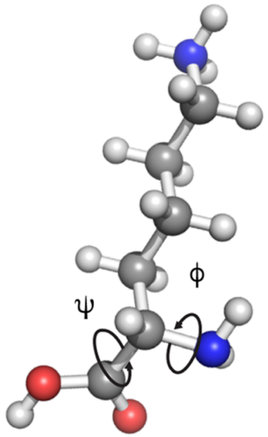
$$E_{\text{dslf_fa13}} = \sum_{\text{S}_1,\text{S}_2} E^{\text{SS}}_{\text{dslf}}(d_{\text{SS}}) + E^{\text{CSS}}_{\text{dslf}}(\theta_{\text{C}_{\beta 1}\text{SS}}) + E^{\text{CSS}}_{\text{dslf}}(\theta_{\text{C}_{\beta 2}\text{SS}}) + E^{\text{C}_\alpha\text{C}_\beta\text{SS}}_{\text{dslf}}(\phi_{\text{C}_{\alpha 1}\text{C}_{\beta 1}\text{SS}}) + E^{\text{C}_\alpha\text{C}_\beta\text{SS}}_{\text{dslf}}(\phi_{\text{C}_{\alpha 2}\text{C}_{\beta 2}\text{SS}}) + E^{\text{C}_\beta\text{SSC}_\beta}_{\text{dslf}}(\phi_{\text{C}_{\beta 1}\text{SSC}_{\beta 2}})$$

i.iii More on Rosetta energy

Alford, R. F., Leaver-Fay, A., Jeliazkov, J. R., O'Meara, M. J., DiMaio, F. P., Park, H., et al. (2017). The Rosetta all-atom energy function for macromolecular modeling and design. Journal of Chemical Theory and Computation, 13(6), 3031–3048. https://doi.org/10.1021/acs.jctc.7b00125

Rosetta's energy on backbone and side-chain torsions

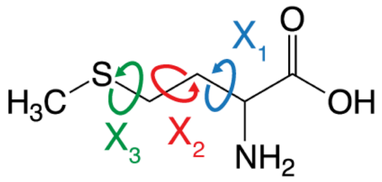
Backbone torsion energies



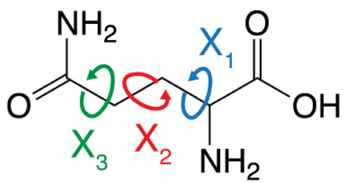
$$E_{\text{rama_prepro}} = \sum_i \begin{cases} -\ln[P_{\text{reg}}(\phi_i, \psi_i | \text{aa}_i)] & \text{C-terminus, } i+1 \neq \text{Pro} \\ -\ln[P_{\text{prepro}}(\phi_i, \psi_i | \text{aa}_i)] & i+1 = \text{Pro} \end{cases}$$

Energies for side-chain rotamer conformations

rotameric



semi-rotameric



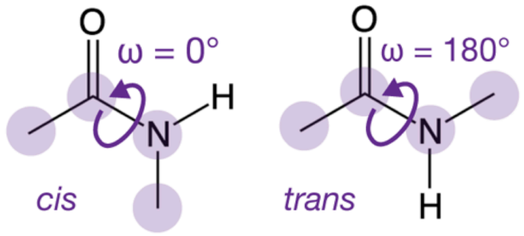
$$E_{\text{fa_dun}} = \sum_r -\ln[P(\text{rot}_r | \phi_r, \psi_r, \text{aa}_r)] + \sum_{k<T_r} \frac{1}{2}\left(\frac{\chi_{k,r} - \mu_{\chi_k}(\phi_r, \psi_r | \text{rot}_r, \text{aa}_r)}{\sigma_{\chi_k}(\phi_r, \psi_r | \text{rot}_r, \text{aa}_r)}\right)^2 - \ln[P(\chi_{T_r,r} | \phi_r, \psi_r, \text{rot}_r, \text{aa}_r)]$$

i.iv More on Rosetta energy

Alford, R. F., Leaver-Fay, A., Jeliazkov, J. R., O'Meara, M. J., DiMaio, F. P., Park, H., et al. (2017). The Rosetta all-atom energy function for macromolecular modeling and design. Journal of Chemical Theory and Computation, 13(6), 3031–3048. https://doi.org/10.1021/acs.jctc.7b00125

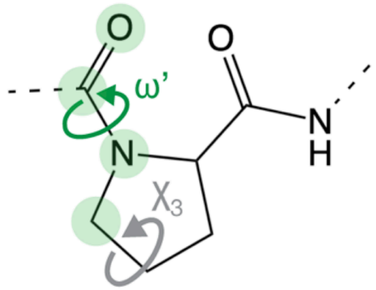
Rosetta's energy on special case torsions.

omega



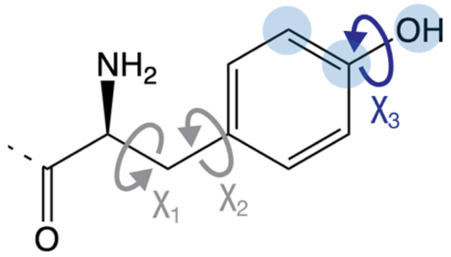
$$E_{\text{omega}} = \sum_r \ln\left(\frac{1}{6\sqrt{2\pi}}\right) - \ln\left(\frac{1}{\sigma_\omega(\phi_r, \psi_r|\text{aa}_r)\sqrt{2\pi}}\right) + \frac{[\omega_r - \mu_\omega(\phi_r, \psi_r|\text{aa}_r)]^2}{2\sigma_\omega^2(\phi_r, \psi_r|\text{aa}_r)}$$

pro_close



$$E_{\text{pro_close}} = \sum_{r \in \text{Pro}} \begin{cases} \dfrac{(\omega_r' - \mu_{\omega'})^2}{\sigma_{\omega'}^2} + \dfrac{\|\text{N}_r - \text{N}_{\text{v},r}\|^2}{\sigma_{\text{N,N}_\text{v}}^2} & r \text{ is not the N-terminus} \\ \dfrac{\|\text{N}_r - \text{N}_{\text{v},r}\|^2}{\sigma_{\text{N,N}_\text{v}}^2} & r \text{ is the N-terminus} \end{cases}$$

yhh_planarity

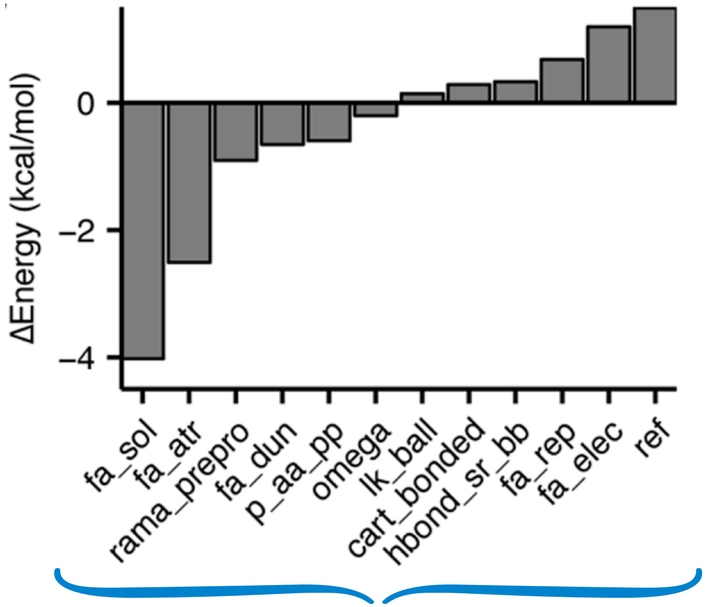


$$E_{\text{yhh_planarity}} = \sum_i \frac{1}{2}[\cos(\pi - 2\chi_{3,i}) + 1]$$

i.v More on Rosetta energy

Alford, R. F., Leaver-Fay, A., Jeliazkov, J. R., O'Meara, M. J., DiMaio, F. P., Park, H., et al. (2017). The Rosetta all-atom energy function for macromolecular modeling and design. Journal of Chemical Theory and Computation, 13(6), 3031–3048. https://doi.org/10.1021/acs.jctc.7b00125

ΔΔG of Mutation:



The reference energy is an empirical baseline in Rosetta, while ΔΔG reflects true stability changes; the reference term affects ΔΔG only indirectly via its contribution to the total energy score.

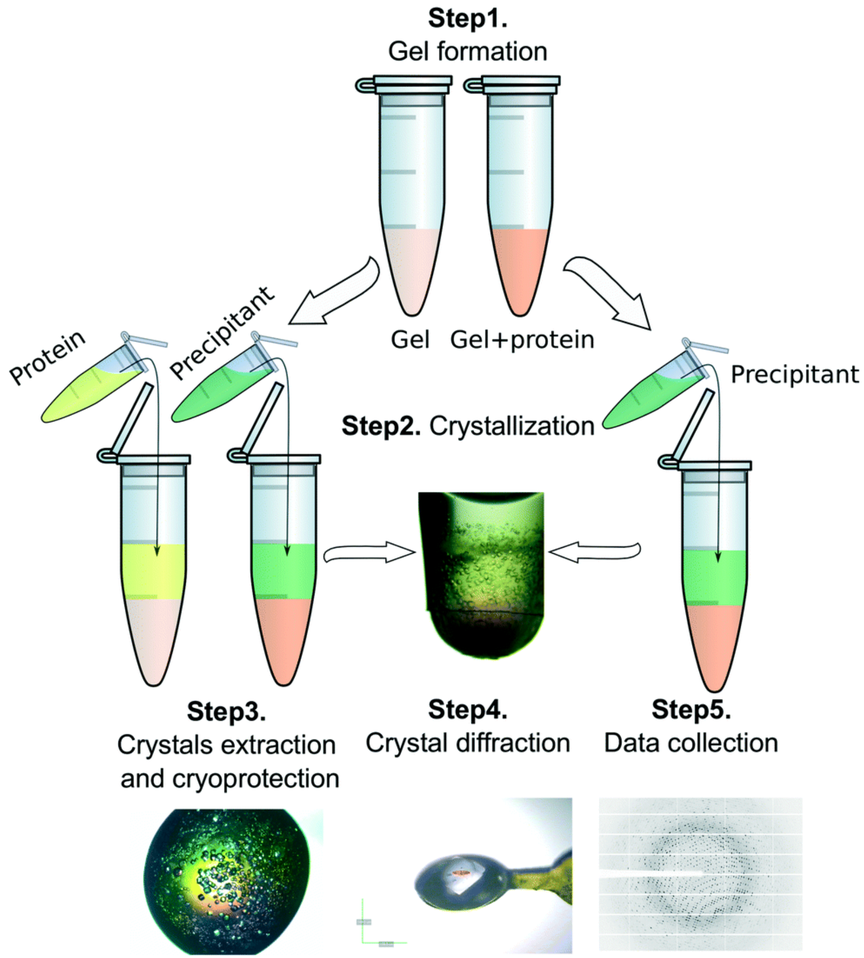
$$E_{\text{ref}} = \sum_i \Delta G_i^{\text{ref}}(\text{aa}_i)$$

Energy Terms (E_i):

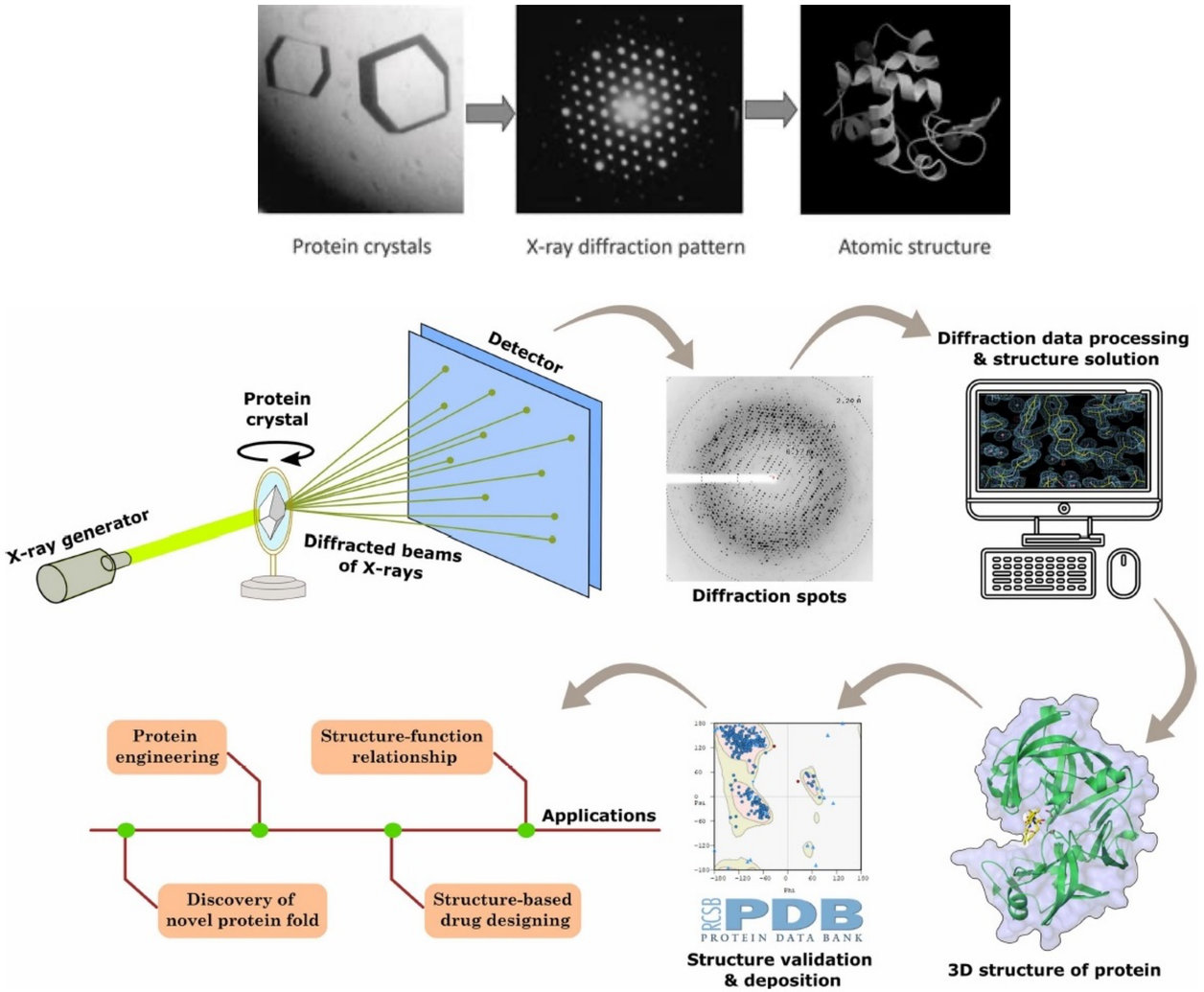
1. Atom-Pair Interactions
2. Protein Backbone and Side-Chain Torsions
3. Special-Case Torsions
4. Modeling Non-ideal Bond Lengths and Angles
5. Protein Design (Reference Energy)

ii. Physical & chemical computations (B-factor & SASA)

What are B-factor, SASA, and how they're related to theromokinetics?
First of all, proteins (e.g., enzymes) can be crystallized.



Standard workflow to prepare protein X-ray crystallography.



X-ray crystallography pipeline and its applications.

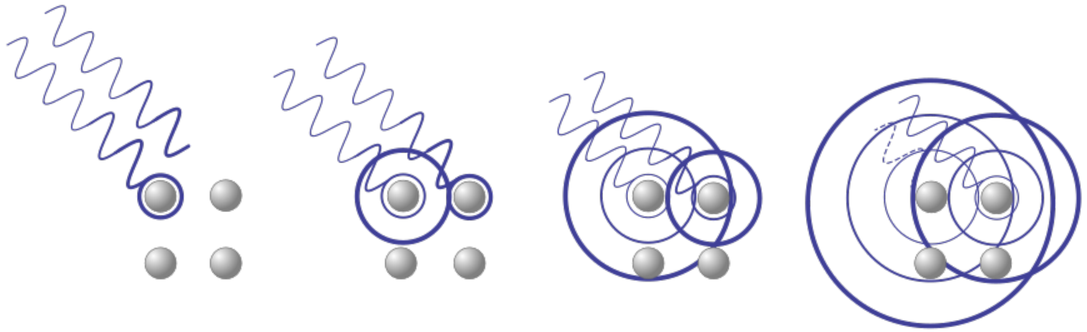
Bhaumik, P. (n.d.). Prasanta Bhaumik's Homepage. Department of Biosciences and Bioengineering, IIT Bombay. Retrieved May 23, 2024, from *https://www.bio.iitb.ac.in/~pbhaumik/*

AR Inspired Pencil. (n.d.). Protein crystallization process [Image]. Retrieved May 23, 2024, from *https://ar.inspiredpencil.com/pictures-2023/protein-crystallization-process*

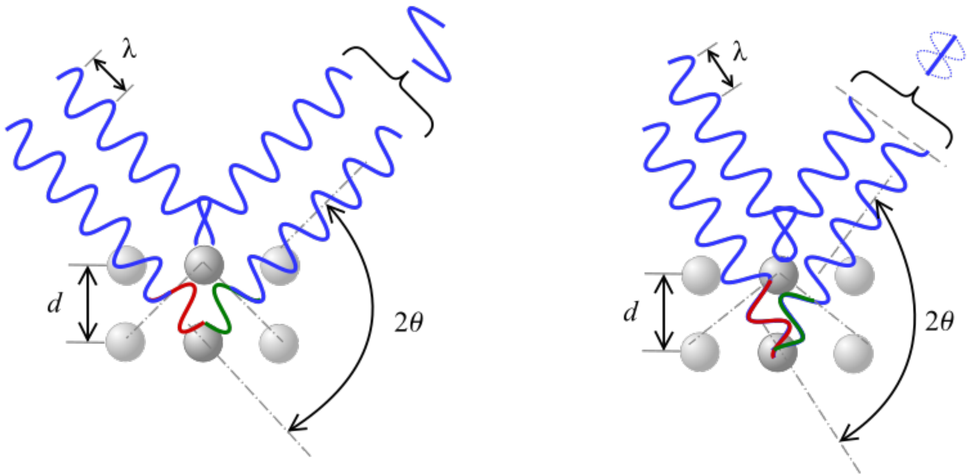
ii. I Bragg’s law & crystallography

Wikipedia contributors. (n.d.). Bragg's law. In Wikipedia. Retrieved May 23, 2024, from *https://en.wikipedia.org/wiki/Bragg%27s_law*

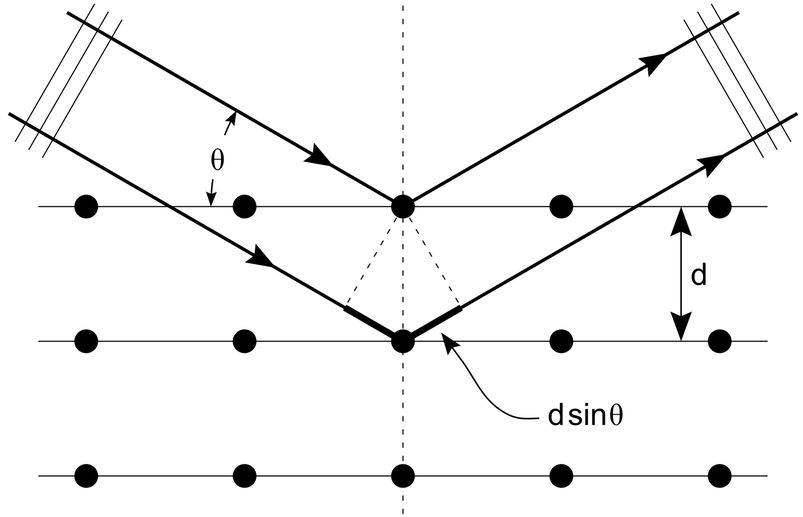
Crystals, at certain specific wavelengths and incident angles, produced intense peaks of reflected radiation.



X-rays interact with the atoms in a crystal.



According to the 2θ deviation, the phase shift causes constructive (left figure) or destructive (right figure) interferences.



$$n\lambda = 2d\sin\theta$$ * (Bragg’s Law)

Constructive interference (Bragg diffraction) occurs when X-rays scattering off crystal planes satisfy Bragg's Law, where the path difference is an integer multiple of the wavelength.

* where λ is the wavelength, n is the diffraction order, θ is the glancing angle, and d is the glancing constant.

ii. ii From Bragg's law to B-factors

Wikipedia contributors. (n.d.). Debye–Waller factor. In Wikipedia. Retrieved May 23, 2024, from *https://en.wikipedia.org/wiki/Debye%E2%80%93Waller_factor*

B-factor is quantified from how much atomic "shaking" reduces diffraction intensities via the Debye-Waller factor.

Derivations:

Step 1. The Debye-Waller factor describes how atomic thermal motion reduces scattering:

$$f_{DW} = \langle \exp(i\mathbf{s}\cdot\mathbf{u})\rangle \approx 1 + i\mathbf{s}\cdot\langle\mathbf{u}\rangle - \frac{1}{2}\langle(\mathbf{s}\cdot\mathbf{u})^2\rangle + \cdots \tag{1}$$

where $\mathbf{s}$ is the scattering vector ($|\mathbf{s}| = 2\sin\theta/\lambda$); $\mathbf{u}$ is the atomic displacement from mean position; $\langle\cdot\rangle$ is the thermal average.

Step 2. For isotropic thermal motion:

i. $\langle\mathbf{u}\rangle = 0$ (no net displacement)

ii. $\langle u_x^2\rangle = \langle u_y^2\rangle = \langle u_z^2\rangle = \langle u^2\rangle/3$

Then:

$$\langle(\mathbf{s}\cdot\mathbf{u})^2\rangle = s^2\langle u^2\rangle\langle\cos^2\phi\rangle = s^2\langle u^2\rangle\cdot\frac{1}{3} \tag{2}$$

Step 3. We then get a Gaussian distribution:

$$\langle \exp(i\mathbf{s}\cdot\mathbf{u})\rangle = \exp\left(-\frac{1}{6}s^2\langle u^2\rangle\right) \tag{3}$$

Step 4. With s inserted, we have:

$$f_{DW} = \exp\left(-\frac{1}{6}\cdot\frac{4\sin^2\theta}{\lambda^2}\cdot\langle u^2\rangle\right) = \exp\left(-\frac{2}{3}\cdot\frac{\sin^2\theta}{\lambda^2}\cdot\langle u^2\rangle\right) \tag{4}$$

Step 5. By convention, we define the *B-factor* as:

$$\boxed{B = 8\pi^2\langle u^2\rangle} \tag{*}$$

B-factor

Substituting $\langle u^2\rangle = B/8\pi^2$:

$$f_{DW} = \exp\left(-\frac{2}{3}\cdot\frac{\sin^2\theta}{\lambda^2}\cdot\frac{B}{8\pi^2}\right) = \exp\left(-\frac{B}{12\pi^2}\cdot\frac{\sin^2\theta}{\lambda^2}\right) \tag{5}$$

Step 6. The standard form used in crystallography is:

$$\boxed{f_{DW} = \exp\left[-B\cdot\frac{\sin^2\theta}{\lambda^2}\right]} \tag{**}$$

Debye-Waller factor

ii. iii Solvent accessible surface area (SASA)

The accessible surface area (ASA) or solvent-accessible surface area (SASA) is the surface area of a biomolecule that is accessible to a solvent.

*SASA Definitions (for more details, check *):*

i. Lee & Richards method (Rolling Probe):

$$\text{SASA} = \text{Area traced by center of solvent sphere}$$

ii. Shrake & Rupley algorithm:

$$\text{SASA} \approx \sum(\text{accessible surface points}) \times (\text{area per point})$$

iii. ΔSASA upon binding:

$$\Delta\text{SASA} = \text{SASA}_{\text{complex}} - (\text{SASA}_{\text{protein}} + \text{SASA}_{\text{ligand}})$$

where negative values indicate burial upon binding.

iv. Hydrophobic effect & folding:

$$\Delta G_{\text{folding}} \propto \Delta\text{SASA}_{\text{hydrophobic}} \times \gamma$$

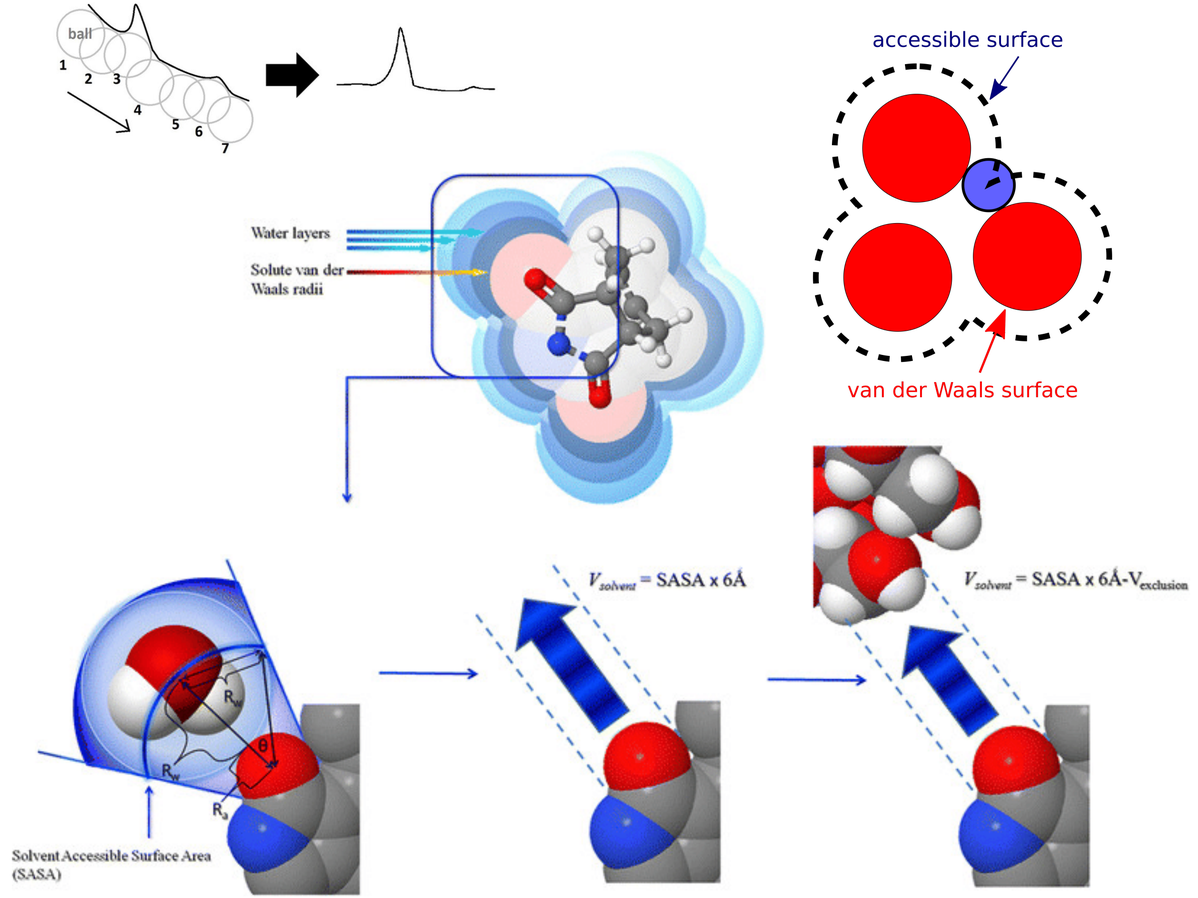
where $\gamma \approx 20 - 25\ \text{cal} \cdot \text{mol}^{-1} \cdot \text{Å}^{-2}$ (hydrophobic energy density).

v. Transfer free energy:

$$\Delta G_{\text{transfer}} = \sum \sigma_i \cdot \text{SASA}_i$$

where σ_i = atomic solvation parameters.

Schematic of the rolling ball algorithm & molecular motions.



* ScienceDirect. (n.d.). Accessible surface area. In ScienceDirect Topics. Elsevier. Retrieved May 23, 2024, from *https://www.sciencedirect.com/topics/biochemistry-genetics-and-molecular-biology/accessible-surface-area*

Niskanen, J., & Laine, S. (2017). A novel method to remove the background from X-ray diffraction signal. Journal of Applied Crystallography, 50(6), 1860–1863. *https://doi.org/10.1107/S1600576717015234*

Liu, S., Cai, C., Tsai, H., Cao, L., Tan, Y., & Gervasio, F. L. (2017). On the fly estimation of host–guest binding free energies using the movable type method: participation in the SAMPL5 blind challenge. *Journal of Computer-Aided Molecular Design, 31*(1), 47–60. *https://doi.org/10.1007/s10822-016-9930-3*

Wikipedia contributors. (n.d.). Accessible surface area. In Wikipedia. Retrieved May 23, 2024, from *https://en.wikipedia.org/wiki/Accessible_surface_area*

ii. iv SASA scores & normalization through BLOSUM

Wikipedia contributors. (2024). BLOSUM. In Wikipedia, The Free Encyclopedia. Retrieved January XX, 2025, from https://en.wikipedia.org/wiki/BLOSUM

Henikoff, S., & Henikoff, J. G. (1992). Amino acid substitution matrices from protein blocks. Proceedings of the National Academy of Sciences, 89(22), 10915–10919. https://doi.org/10.1073/pnas.89.22.10915

SASA is computed from the predicted 3D structure for WT and mutant, and then normalized using BLOSUM scores to scale structural changes by the evolutionary conservativeness of the amino-acid substitution.

	C	S	T	A	G	P	D	E	Q	N	H	R	K	M	I	L	V	W	Y	F	
C	9																				C
S	-1	4																			S
T	-1	1	5																		T
A	0	1	0	4																	A
G	-3	0	-2	0	6																G
P	-3	-1	-1	-1	-2	7															P
D	-3	0	-1	-2	-1	-1	6														D
E	-4	0	-1	-1	-2	-1	2	5													E
Q	-3	0	-1	-1	-2	-1	0	2	5												Q
N	-3	1	0	-2	0	-2	1	0	0	6											N
H	-3	-1	-2	-2	-2	-2	-1	0	0	1	8										H
R	-3	-1	-1	-1	-2	-2	-2	0	1	0	0	5									R
K	-3	0	-1	-1	-2	-1	-1	1	1	0	-1	2	5								K
M	-1	-1	-1	-1	-3	-2	-3	-2	0	-2	-2	-1	-1	5							M
I	-1	-2	-1	-1	-4	-3	-3	-3	-3	-3	-3	-3	-3	1	4						I
L	-1	-2	-1	-1	-4	-3	-4	-3	-2	-3	-3	-2	-2	2	2	4					L
V	-1	-2	0	0	-3	-2	-3	-2	-2	-3	-3	-3	-2	1	3	1	4				V
W	-2	-3	-2	-3	-2	-4	-4	-3	-2	-4	-2	-3	-3	-1	-3	-2	-3	11			W
Y	-2	-2	-2	-2	-3	-3	-3	-2	-1	-2	2	-2	-2	-1	-1	-1	-1	2	7		Y
F	-2	-2	-2	-2	-3	-4	-3	-3	-3	-3	-1	-3	-3	0	0	0	-1	1	3	6	F
	C	S	T	A	G	P	D	E	Q	N	H	R	K	M	I	L	V	W	Y	F	

The BLOSUM62 matrix, the amino acids have been grouped and coloured based on Dayhoff's classification scheme (e.g. hydrophobic, polar, acidic, basic, aromatic, etc.).

Calculating Frequency & Probability:

$$LogOddRatio = 2\log_2\left(\frac{P(O)}{P(E)}\right)$$

Calculating BLOSUM Matrices:

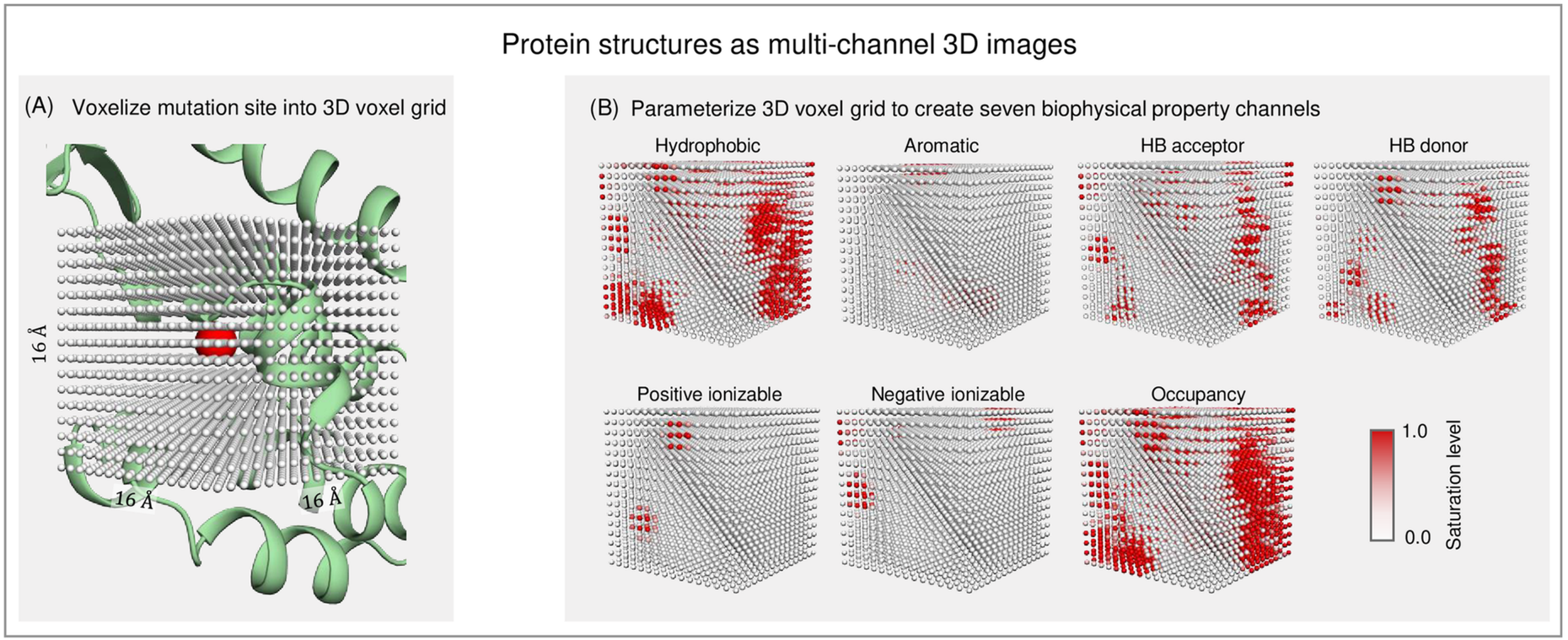
$$S_{ij} = \frac{1}{\lambda}\log\frac{p_{ij}}{q_i q_j}$$

i. If BLOSUM is high → mutation is expected → SASA change should be less penalized
ii. If BLOSUM is low → mutation is rare/disruptive → SASA change should be amplified

iii. Predicted 3D structure (3D CNN)

Li, B., Yang, Y. T., Capra, J. A., & Gerstein, M. B. (2020). Predicting changes in protein thermodynamic stability upon point mutation with deep 3D convolutional neural networks. PLOS Computational Biology, 16(11), e1008291. https://doi.org/10.1371/journal.pcbi.1008291

ThermoNet shows that treating protein structures as 3D images, and analyzing mutation environments with a 3D CNN, which is an effective way to predict stability changes upon mutation.

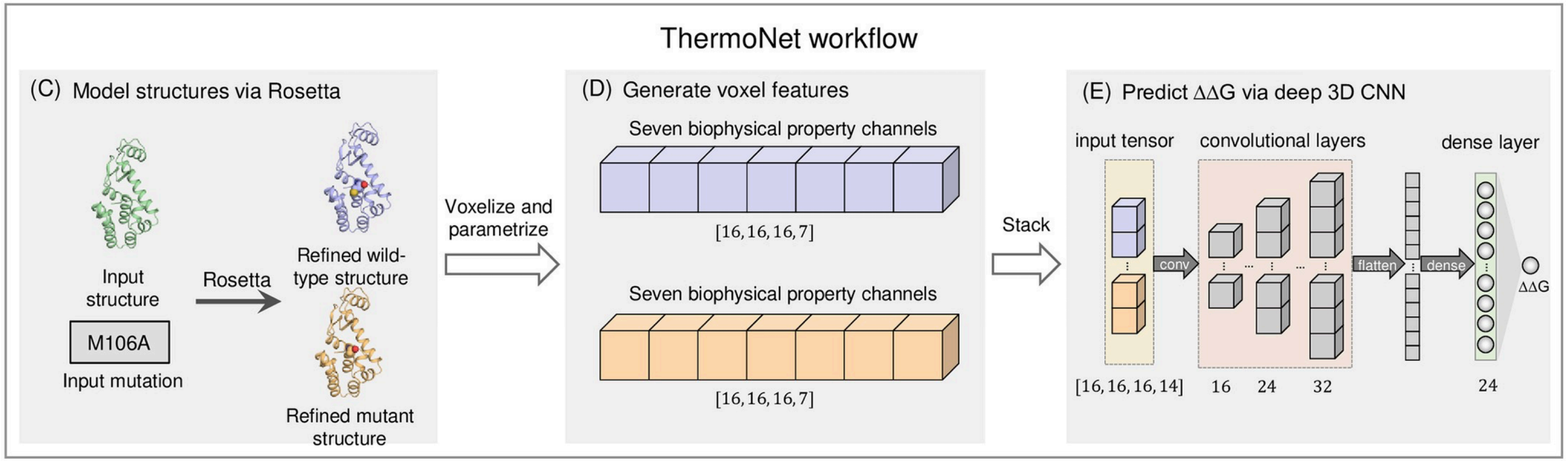


A 16 Å cubic region around the mutation site is voxelized at 1 Å resolution and encoded with seven biophysical property channels (hydrophobic, aromatic, donor, acceptor, positive, negative, occupancy).

iii. Predicted 3D structure (3D CNN)

Li, B., Yang, Y. T., Capra, J. A., & Gerstein, M. B. (2020). Predicting changes in protein thermodynamic stability upon point mutation with deep 3D convolutional neural networks. PLOS Computational Biology, 16(11), e1008291. https://doi.org/10.1371/journal.pcbi.1008291

Rosetta-refined wild-type and mutant structures are voxelized around the mutation site into two [16,16,16,7] feature maps, stacked into a [16,16,16,14] tensor, and passed to a 3D CNN to predict ΔΔG.



The 3D CNN outputs the predicted ΔΔG caused by the mutation:

$$\Delta\Delta G = \Delta G_{mutant} - \Delta G_{wild-type} = -RTln\frac{[folded]_{mutant}/[unfolded]_{mutant}}{[folded]_{wild-type}/[unfolded]_{wild-type}}$$

iv. Ensemble: physical-chemical models & 3D-CNN

Deep Learning Model

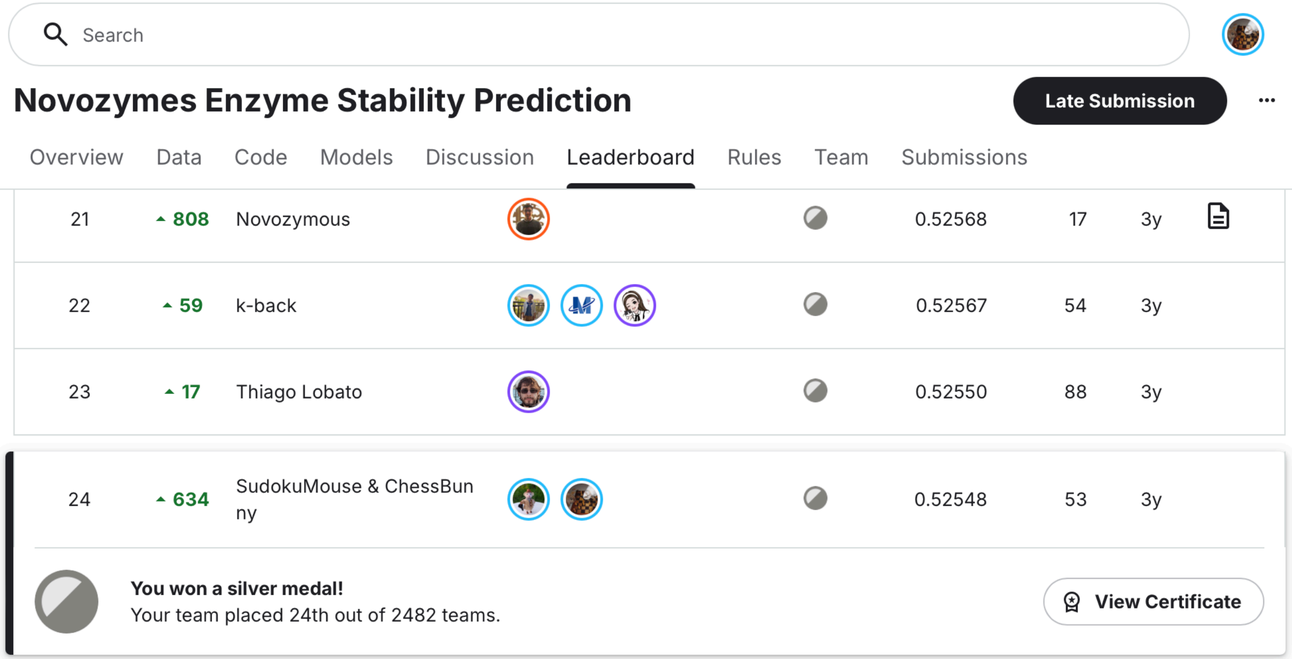
Physical-chemical models

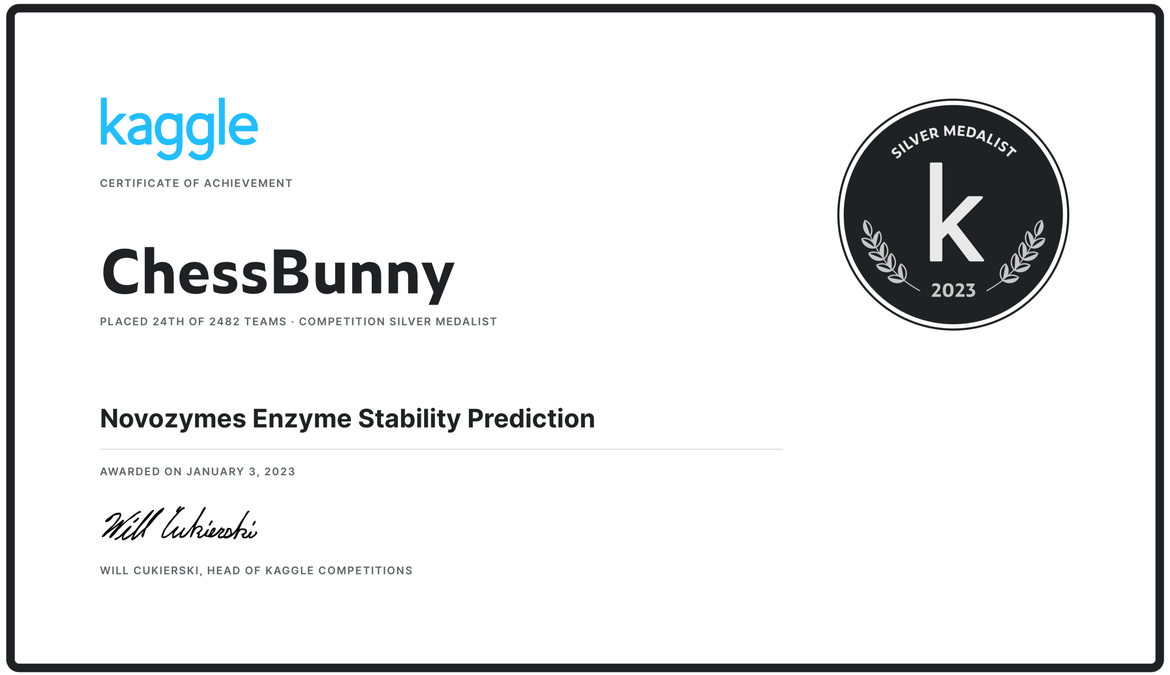
$$\text{Ensemble} = \frac{w_1*[\text{B-factor}] + w_2*[\text{SASA}*\text{BLOSUM62}] + w_3*[\text{3D-CNN}]}{3}$$

Results

Results

We won a silver medal!

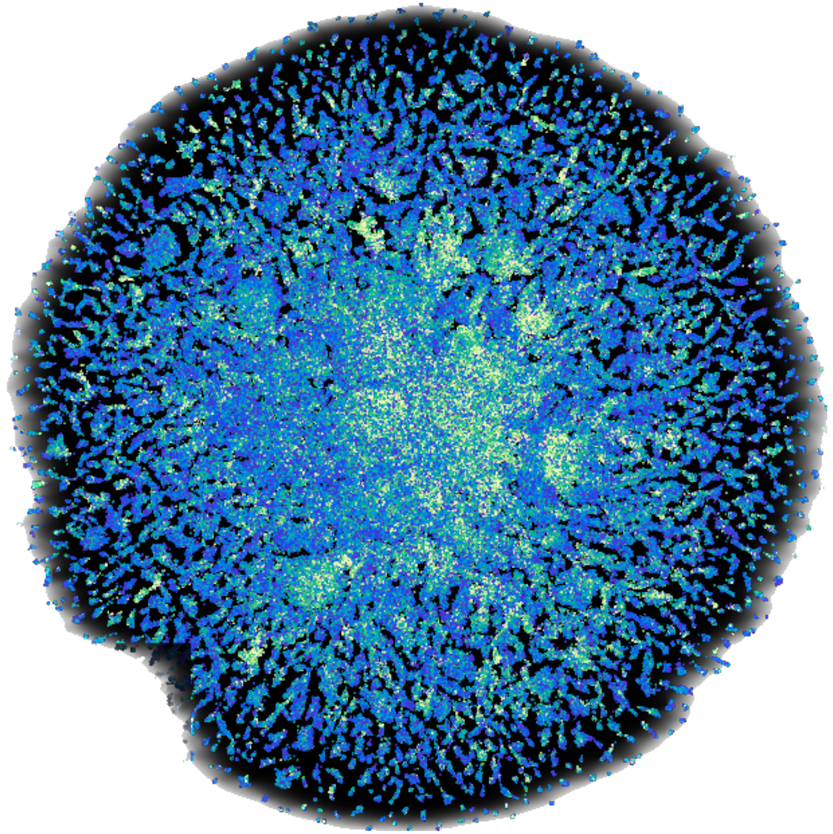




(Yes, I call myself chessbunny.)

Conclusions

Conclusions & Recommendations



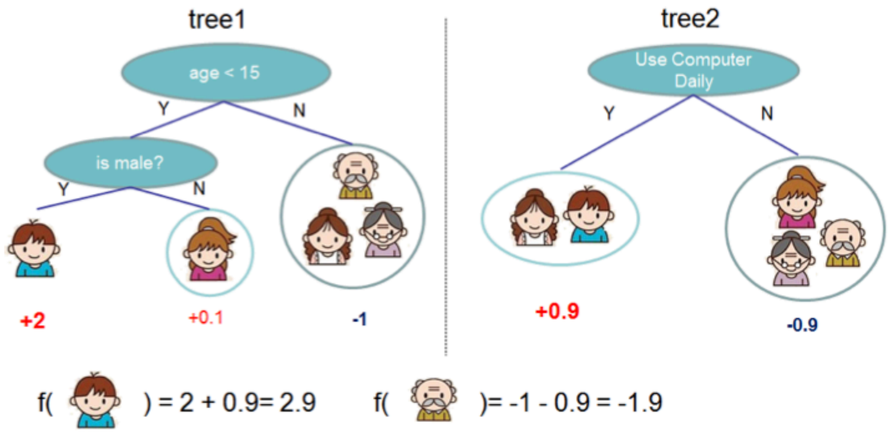
Exploring 1 million out of 772M proteins (ESM).

AI is revolutionizing our ability to predict protein stability, opening up incredible new possibilities for discovery. By embracing its power while being mindful of its current limitations, we can guide it to even greater breakthroughs.

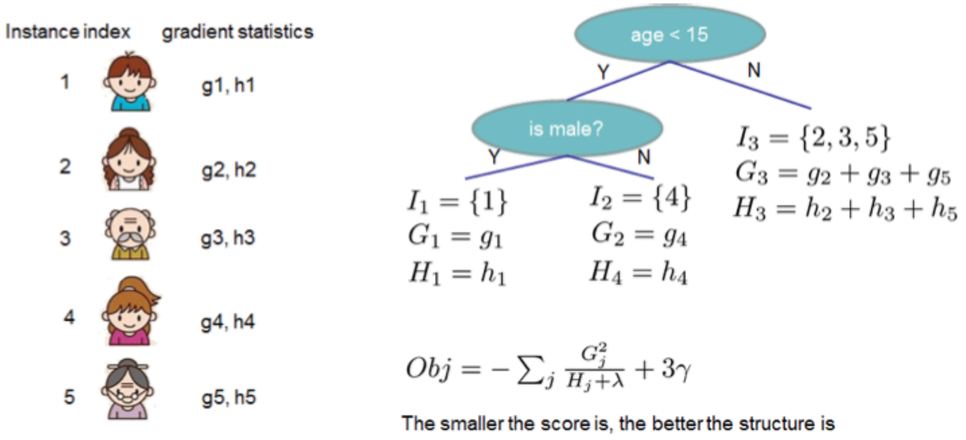
Appendix

i. Extreme Gradient Boosting (XGBoost) Model

*Chen, T., & Guestrin, C. (2016). XGBoost: A Scalable Tree Boosting System. In Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (pp. 785-794). ACM. *https://doi.org/10.1145/2939672.2939785*



Tree Ensemble Model. The final prediction for a given example is the sum of predictions from each tree.



Structure Score Calculation. We only need to sum up the gradient and second order gradient statistics on each leaf, then apply the scoring formula to get the quality score.

We minimize the objective function by sequentially adding trees that optimally reduce loss while controlling complexity.

*Derivations (for more details, check *):*

Step 1. Main objective function:

$$\mathcal{L}(\phi) = \sum_i l(\hat{y}_i, y_i) + \sum_k \Omega(f_k) \qquad (*)$$

where

$$\Omega(f) = \gamma T + \frac{1}{2}\lambda \|w\|^2$$

Step 2. At iteration t (remove the constant terms):

$$\tilde{\mathcal{L}}^{(t)} = \sum_{i=1}^{n} \left[g_i f_t(\mathbf{x}_i) + \frac{1}{2} h_i f_t^2(\mathbf{x}_i) \right] + \Omega(f_t) \qquad (1)$$

Step 3. Define $I_j = \{i | q(\mathbf{x}_i) = j\}$ as the instance set of leaf j. We can rewrite Eq (1) by expanding Ω as follows:

$$\begin{aligned} \tilde{\mathcal{L}}^{(t)} &= \sum_{i=1}^{n} \left[g_i f_t(\mathbf{x}_i) + \frac{1}{2} h_i f_t^2(\mathbf{x}_i) \right] + \gamma T + \frac{1}{2}\lambda \sum_{j=1}^{T} w_j^2 \\ &= \sum_{j=1}^{T} \left[\left(\sum_{i \in I_j} g_i \right) w_j + \frac{1}{2} \left(\sum_{i \in I_j} h_i + \lambda \right) w_j^2 \right] + \gamma T \end{aligned} \qquad (2)$$

Step 4. For a fixed structure $q(\mathbf{x})$, we can compute the optimal weight w_j^* of leaf j by:

$$w_j^* = -\frac{\sum_{i \in I_j} g_i}{\sum_{i \in I_j} h_i + \lambda} \qquad (3)$$

ii. ESM-2

Hsu, C., Nisonoff, H., Fannjiang, C., & Listgarten, J. (2020). Learning inverse folding from millions of predicted structures. BioRxiv. *https://doi.org/10.1101/2020.12.15.422761*

Cheng, J., et al. (2022). ESM Metagenomic Atlas: The first view of the 'dark matter' of the protein universe. BioRxiv. *https://doi.org/10.1101/2022.07.20.500902*

Ramith H. (n.d.). *ESM-2: Evolutionary-scale prediction of atomic-level protein structure with a language model*. Ramith.fyi. Retrieved May 23, 2024, from *https://ramith.fyi/esm-2-evolutionary-scale-prediction-of-atomic-level-protein-structure-with-a-language-model/*

ESM-2 learns protein sequences through masked language modeling, and we discover its attention patterns using logistic regression (*).

(*)

$$p(c_{ij}) = \left(1 + \exp\left(-\beta_0 - \sum_{l=1}^{L}\sum_{k=1}^{K}\beta_{kl}a_{ij}^{kl}\right)\right)^{-1}$$

value of the attention map at i,j

RV {True , False} , which is true if amino acids i, j are in contact

K = # of attention heads

L = # of layers

